

# Quadratics & Nonlinear Equations

## ANSWER KEY



### Solving quadratics by factoring

- 1 Solve  $x^2 - 5x + 6 = 0$  by factoring.

Factor:  $(x - 2)(x - 3) = 0$ , so  $x = 2$  or  $x = 3$ .

- 2 Solve  $2x^2 + 7x - 4 = 0$  by factoring.

Factor:  $(2x - 1)(x + 4) = 0$ , so  $x = \frac{1}{2}$  or  $x = -4$ .

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### Solving quadratics with the quadratic formula

- 3 Solve  $x^2 + 4x - 1 = 0$  using the quadratic formula, giving exact answers.

$$x = \frac{-4 \pm \sqrt{16 + 4}}{2} = \frac{-4 \pm \sqrt{20}}{2} = \frac{-4 \pm 2\sqrt{5}}{2} = -2 \pm \sqrt{5}.$$

- 4 Use the discriminant to determine the number of real solutions of  $3x^2 - 2x + 5 = 0$ , without solving the equation.

Discriminant  $= b^2 - 4ac = (-2)^2 - 4(3)(5) = 4 - 60 = -56$ . Since the discriminant is negative, the equation has no real solutions (two complex solutions).

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### Completing the square and vertex form

- 5 Write  $f(x) = x^2 - 6x + 11$  in vertex form by completing the square, and state the vertex.

$$x^2 - 6x + 11 = (x^2 - 6x + 9) + 2 = (x - 3)^2 + 2. \text{ Vertex form: } f(x) = (x - 3)^2 + 2. \text{ Vertex: } (3, 2).$$

- 6 A parabola has vertex  $(2, -5)$  and passes through  $(4, -1)$ . Find its equation in vertex form.

Vertex form:  $y = a(x - 2)^2 - 5$ . Substitute  $(4, -1)$ :  $-1 = a(4 - 2)^2 - 5$ , so  $-1 = 4a - 5$ , giving  $4a = 4$ ,  $a = 1$ . Equation:  $y = (x - 2)^2 - 5$ .

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### Finding zeros, maximum, and minimum values

- 7 A ball's height is modeled by  $h(t) = -16t^2 + 64t + 5$ , where  $t$  is time in seconds. Find the maximum height and the time at which it occurs.

The maximum occurs at  $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$  seconds. Maximum height:  
 $h(2) = -16(4) + 64(2) + 5 = -64 + 128 + 5 = 69$  feet.

- 8 Find the zeros of  $f(x) = x^2 - 9$ , and interpret them as  $x$ -intercepts on a graph.

$x^2 - 9 = 0$ , so  $x^2 = 9$ , giving  $x = 3$  or  $x = -3$ . These are the two points where the parabola crosses the  $x$ -axis:  $(3, 0)$  and  $(-3, 0)$ .

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## Systems with one linear and one quadratic equation

- 9 Solve the system:  $y = x^2 - 2x - 3$  and  $y = x - 1$ .

Set equal:  $x^2 - 2x - 3 = x - 1$ . Subtract  $x - 1$  from both sides:  $x^2 - 3x - 2 = 0$ . Using the quadratic formula:  $x = \frac{3 \pm \sqrt{9 + 8}}{2} = \frac{3 \pm \sqrt{17}}{2}$ . The corresponding  $y$ -values come from  $y = x - 1$  for each  $x$ -value.

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## Exponential growth and decay

- 10 A population of bacteria triples every hour, starting with 200 bacteria. Write an exponential model and find the population after 4 hours.

Model:  $P(t) = 200(3)^t$ . After 4 hours:  $P(4) = 200(3^4) = 200(81) = 16200$  bacteria.

- 11 A car worth 24000 dollars depreciates by 15 percent each year. Write an exponential model and find its value after 3 years, to the nearest dollar.

Model:  $V(t) = 24000(0.85)^t$ . After 3 years:  $V(3) = 24000(0.85)^3 = 24000(0.614125) \approx 14739$  dollars.

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## Radical and rational equations

- 12 Solve  $\sqrt{2x + 3} = 5$ , and check your solution.

Square both sides:  $2x + 3 = 25$ , so  $2x = 22$ , giving  $x = 11$ . Check:  $\sqrt{2(11) + 3} = \sqrt{25} = 5$  ✓.

- 13 Solve  $\frac{3}{x - 2} = \frac{1}{x + 1}$ , and state any restrictions on  $x$ .

Cross-multiply:  $3(x + 1) = 1(x - 2)$ , so  $3x + 3 = x - 2$ , giving  $2x = -5$ ,  $x = -2.5$ . Restrictions:  $x \neq 2$  and  $x \neq -1$  (values that would make a denominator zero); since  $x = -2.5$  avoids both, it is valid.

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## Nonlinear relationships in context

- 14 The area of a square garden is modeled by  $A(s) = s^2$ , where  $s$  is the side length in meters. If the area must be at most 144 square meters, find the maximum possible side length.

$s^2 \leq 144$ , so  $s \leq 12$  (taking the positive root since side length must be positive). The maximum side length is 12 meters.

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## Factoring special products

- 15 Factor  $9x^2 - 25$  completely.

This is a difference of squares:  $9x^2 - 25 = (3x - 5)(3x + 5)$ .

- 16 Factor  $x^2 + 10x + 25$  completely, and identify the special pattern used.

This is a perfect square trinomial:  $x^2 + 10x + 25 = (x + 5)^2$ .

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## Converting between standard, factored, and vertex forms

- 17 A parabola is given in factored form as  $y = 2(x - 1)(x + 3)$ . Convert this to standard form.

Expand:  $2(x - 1)(x + 3) = 2(x^2 + 3x - x - 3) = 2(x^2 + 2x - 3) = 2x^2 + 4x - 6$ .

- 18 A parabola has zeros at  $x = 2$  and  $x = 6$ , and passes through  $(4, -8)$ . Find its equation in factored form.

Factored form:  $y = a(x - 2)(x - 6)$ . Substitute  $(4, -8)$ :  $-8 = a(4 - 2)(4 - 6) = a(2)(-2) = -4a$ , so  $a = 2$ . Equation:  $y = 2(x - 2)(x - 6)$ .

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## Quadratic word problems: projectile motion

- 19 A rocket is launched with height modeled by  $h(t) = -16t^2 + 96t$ , where  $t$  is time in seconds. Find when the rocket hits the ground, and find its maximum height.

The rocket hits the ground when  $h(t) = 0$ :  $-16t^2 + 96t = 0$ , so  $-16t(t - 6) = 0$ , giving  $t = 0$  or  $t = 6$  seconds (the rocket lands at  $t = 6$ ). Maximum height occurs at  $t = -\frac{96}{2(-16)} = 3$  seconds:  
 $h(3) = -16(9) + 96(3) = -144 + 288 = 144$  feet.

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## Quadratic optimization in business contexts

- 20 A company's revenue is modeled by  $R(x) = -5x^2 + 300x$ , where  $x$  is the price per item in dollars. Find the price that maximizes revenue, and find the maximum revenue.

Maximum occurs at  $x = -\frac{300}{2(-5)} = 30$  dollars. Maximum revenue:  
 $R(30) = -5(900) + 300(30) = -4500 + 9000 = 4500$  dollars.

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## Determining the number of solutions graphically

- 21 A parabola  $y = x^2 - 4x + 7$  and a line  $y = 2x - 5$  are graphed together. Use the discriminant of the resulting equation to determine how many times the graphs intersect.

Set equal:  $x^2 - 4x + 7 = 2x - 5$ , so  $x^2 - 6x + 12 = 0$ . Discriminant:  
 $(-6)^2 - 4(1)(12) = 36 - 48 = -12$ . Since the discriminant is negative, there are no real solutions, meaning the parabola and line never intersect.

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### Multiple choice — easy 15

- 22 Solve  $x^2 + 5x = 0$  by factoring.

- a.  $(-4, 0)$
- b.  $(5, 0)$
- c.  $(-5, 0)$  ✓
- d.  $(-5, -1)$

Factor:  $(x - (0))(x - (-5)) = 0$ , so  $x = 0$  or  $x = -5$ .

- 23 Solve  $x^2 - 1 = 0$  by factoring.

- a.  $(-1, 1)$  ✓
- b.  $(1, -1)$
- c.  $(-1, 0)$
- d.  $(0, 1)$

Factor:  $(x - (-1))(x - (1)) = 0$ , so  $x = -1$  or  $x = 1$ .

- 24 Solve  $x^2 - 7x - 44 = 0$  by factoring.

- a.  $(-4, 11)$  ✓
- b.  $(-4, 10)$
- c.  $(4, -11)$
- d.  $(-3, 11)$

Factor:  $(x - (11))(x - (-4)) = 0$ , so  $x = 11$  or  $x = -4$ .

25 Solve  $x^2 - 10x + 9 = 0$  by factoring.

a. (2, 9)

b. (1, 8)

c. (-1, -9)

d. (1, 9) ✓

Factor:  $(x - (9))(x - (1)) = 0$ , so  $x = 9$  or  $x = 1$ .

26 Solve  $x^2 + 16x + 55 = 0$  by factoring.

a. (-11, -6)

b. (-11, -5) ✓

c. (-10, -5)

d. (11, 5)

Factor:  $(x - (-5))(x - (-11)) = 0$ , so  $x = -5$  or  $x = -11$ .

27 Solve  $x^2 + 14x + 33 = 0$  by factoring.

a. (11, 3)

b. (-11, -3) ✓

c. (-10, -3)

d. (-11, -4)

Factor:  $(x - (-11))(x - (-3)) = 0$ , so  $x = -11$  or  $x = -3$ .

28 Solve  $x^2 - 8x + 15 = 0$  by factoring.

a. (3, 4)

b. (-3, -5)

c. (4, 5)

d. (3, 5) ✓

Factor:  $(x - (3))(x - (5)) = 0$ , so  $x = 3$  or  $x = 5$ .

29 Solve  $x^2 - 9x - 36 = 0$  by factoring.

- a.  $(3, -12)$
- b.  $(-2, 12)$
- c.  $(-3, 12)$  ✓
- d.  $(-3, 11)$

Factor:  $(x - (-3))(x - (12)) = 0$ , so  $x = -3$  or  $x = 12$ .

30 Solve  $x^2 + 8x + 12 = 0$  by factoring.

- a.  $(-5, -2)$
- b.  $(-6, -2)$  ✓
- c.  $(6, 2)$
- d.  $(-6, -3)$

Factor:  $(x - (-2))(x - (-6)) = 0$ , so  $x = -2$  or  $x = -6$ .

31 Solve  $x^2 + 8x - 48 = 0$  by factoring.

- a.  $(-12, 4)$  ✓
- b.  $(-11, 4)$
- c.  $(-12, 3)$
- d.  $(12, -4)$

Factor:  $(x - (4))(x - (-12)) = 0$ , so  $x = 4$  or  $x = -12$ .

32 Solve  $x^2 - 3x = 0$  by factoring.

a. (1, 3)

b. (0, 3) ✓

c. (0, -3)

d. (0, 2)

Factor:  $(x - (3))(x - (0)) = 0$ , so  $x = 3$  or  $x = 0$ .

33 Solve  $x^2 - 49 = 0$  by factoring.

a. (-6, 7)

b. (-7, 7) ✓

c. (-7, 6)

d. (7, -7)

Factor:  $(x - (-7))(x - (7)) = 0$ , so  $x = -7$  or  $x = 7$ .

34 Solve  $x^2 - 8x + 12 = 0$  by factoring.

a. (2, 5)

b. (3, 6)

c. (-2, -6)

d. (2, 6) ✓

Factor:  $(x - (6))(x - (2)) = 0$ , so  $x = 6$  or  $x = 2$ .

35 Solve  $x^2 + 12x + 11 = 0$  by factoring.

a. (-11, -1) ✓

b. (-11, -2)

c. (11, 1)

d. (-10, -1)

Factor:  $(x - (-1))(x - (-11)) = 0$ , so  $x = -1$  or  $x = -11$ .

36 Solve  $x^2 - 16 = 0$  by factoring.

a.  $(-4, 4)$  ✓

b.  $(-4, 3)$

c.  $(4, -4)$

d.  $(-3, 4)$

Factor:  $(x - (4))(x - (-4)) = 0$ , so  $x = 4$  or  $x = -4$ .

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### Multiple choice — medium 15

37 Find the vertex of the parabola  $f(x) = 2x^2 - 4x - 6$ .

a.  $(1, -8)$  ✓

b.  $(-1, -8)$

c.  $(2, -8)$

d.  $(1, 8)$

The vertex occurs at  $x = -b/(2a) = -(-4)/(2(2)) = 1$ .  $f(1) = -8$ . Vertex:  $(1, -8)$ .

38 Find the vertex of the parabola  $f(x) = -3x^2 - 60x - 6$ .

a.  $(-9, 294)$

b.  $(-10, -294)$

c.  $(10, 294)$

d.  $(-10, 294)$  ✓

The vertex occurs at  $x = -b/(2a) = -(-60)/(2(-3)) = -10$ .  $f(-10) = 294$ . Vertex:  $(-10, 294)$ .

39 Find the vertex of the parabola  $f(x) = x^2 - 8x - 6$ .

- a.  $(4, -22)$  ✓
- b.  $(5, -22)$
- c.  $(4, 22)$
- d.  $(-4, -22)$

The vertex occurs at  $x = -b/(2a) = -(-8)/(2(1)) = 4$ .  $f(4) = -22$ . Vertex:  $(4, -22)$ .

40 Find the vertex of the parabola  $f(x) = -x^2 - 12x - 5$ .

- a.  $(-5, 31)$
- b.  $(-6, -31)$
- c.  $(-6, 31)$  ✓
- d.  $(6, 31)$

The vertex occurs at  $x = -b/(2a) = -(-12)/(2(-1)) = -6$ .  $f(-6) = 31$ . Vertex:  $(-6, 31)$ .

41 Find the vertex of the parabola  $f(x) = -3x^2 - 18x - 9$ .

- a.  $(-2, 18)$
- b.  $(-3, -18)$
- c.  $(-3, 18)$  ✓
- d.  $(3, 18)$

The vertex occurs at  $x = -b/(2a) = -(-18)/(2(-3)) = -3$ .  $f(-3) = 18$ . Vertex:  $(-3, 18)$ .

42 Find the vertex of the parabola  $f(x) = x^2 + 16x + 6$ .

- a.  $(-8, -58)$  ✓
- b.  $(8, -58)$
- c.  $(-7, -58)$
- d.  $(-8, 58)$

The vertex occurs at  $x = -b/(2a) = -(16)/(2(1)) = -8$ .  $f(-8) = -58$ . Vertex:  $(-8, -58)$ .

43 Find the vertex of the parabola  $f(x) = -x^2 + 2x - 2$ .

a.  $(-1, -1)$

b.  $(1, -1)$  ✓

c.  $(2, -1)$

d.  $(1, 1)$

The vertex occurs at  $x = -b/(2a) = -(2)/(2(-1)) = 1$ .  $f(1) = -1$ . Vertex:  $(1, -1)$ .

44 Find the vertex of the parabola  $f(x) = 3x^2 - 36x - 7$ .

a.  $(6, 115)$

b.  $(7, -115)$

c.  $(6, -115)$  ✓

d.  $(-6, -115)$

The vertex occurs at  $x = -b/(2a) = -(-36)/(2(3)) = 6$ .  $f(6) = -115$ . Vertex:  $(6, -115)$ .

45 Find the vertex of the parabola  $f(x) = -3x^2 - 18x - 15$ .

a.  $(-3, 12)$  ✓

b.  $(3, 12)$

c.  $(-3, -12)$

d.  $(-2, 12)$

The vertex occurs at  $x = -b/(2a) = -(-18)/(2(-3)) = -3$ .  $f(-3) = 12$ . Vertex:  $(-3, 12)$ .

46 Find the vertex of the parabola  $f(x) = -3x^2 - 60x - 15$ .

a.  $(-10, 285)$  ✓

b.  $(10, 285)$

c.  $(-10, -285)$

d.  $(-9, 285)$

The vertex occurs at  $x = -b/(2a) = -(-60)/(2(-3)) = -10$ .  $f(-10) = 285$ . Vertex:  $(-10, 285)$ .

47 Find the vertex of the parabola  $f(x) = -x^2 + 6x - 11$ .

a.  $(-3, -2)$

b.  $(4, -2)$

c.  $(3, -2)$  ✓

d.  $(3, 2)$

The vertex occurs at  $x = -b/(2a) = -(6)/(2(-1)) = 3$ .  $f(3) = -2$ . Vertex:  $(3, -2)$ .

48 Find the vertex of the parabola  $f(x) = x^2 - 8x - 10$ .

a.  $(-4, -26)$

b.  $(5, -26)$

c.  $(4, -26)$  ✓

d.  $(4, 26)$

The vertex occurs at  $x = -b/(2a) = -(-8)/(2(1)) = 4$ .  $f(4) = -26$ . Vertex:  $(4, -26)$ .

49 Find the vertex of the parabola  $f(x) = x^2 - 14x - 2$ .

a.  $(-7, -51)$

b.  $(7, 51)$

c.  $(7, -51)$  ✓

d.  $(8, -51)$

The vertex occurs at  $x = -b/(2a) = -(-14)/(2(1)) = 7$ .  $f(7) = -51$ . Vertex:  $(7, -51)$ .

50 Find the vertex of the parabola  $f(x) = -x^2 + 6x + 11$ .

a.  $(3, -20)$

b.  $(-3, 20)$

c.  $(3, 20)$  ✓

d.  $(4, 20)$

The vertex occurs at  $x = -b/(2a) = -(6)/(2(-1)) = 3$ .  $f(3) = 20$ . Vertex:  $(3, 20)$ .

51 Find the vertex of the parabola  $f(x) = -x^2 - 12x - 10$ .

a.  $(-5, 26)$

b.  $(-6, -26)$

c.  $(-6, 26)$  ✓

d.  $(6, 26)$

The vertex occurs at  $x = -b/(2a) = -(-12)/(2(-1)) = -6$ .  $f(-6) = 26$ . Vertex:  $(-6, 26)$ .

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### Multiple choice — hard 15

52 A rocket is launched with height modeled by  $h(t) = -16t^2 + 288t$ , where  $t$  is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).

a.  $(10, 1296)$

b.  $(9, 1312)$

c.  $(9, 1296)$  ✓

d.  $(18, 1296)$

Maximum occurs at  $t = -288/(2(-16)) = 9$  seconds.  $h(9) = -16(9)^2 + 288(9) = 1296$ .  $(9, 1296)$ .

- 53 A rocket is launched with height modeled by  $h(t) = -16t^2 + 64t$ , where  $t$  is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).
- a. (4, 64)
  - b. (2, 80)
  - c. (2, 64) ✓
  - d. (3, 64)

Maximum occurs at  $t = -64/(2(-16)) = 2$  seconds.  $h(2) = -16(2)^2 + 64(2) = 64$ . (2, 64).

- 54 A rocket is launched with height modeled by  $h(t) = -16t^2 + 128t$ , where  $t$  is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).
- a. (4, 256) ✓
  - b. (4, 272)
  - c. (5, 256)
  - d. (8, 256)

Maximum occurs at  $t = -128/(2(-16)) = 4$  seconds.  $h(4) = -16(4)^2 + 128(4) = 256$ . (4, 256).

- 55 A rocket is launched with height modeled by  $h(t) = -16t^2 + 256t$ , where  $t$  is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).
- a. (9, 1024)
  - b. (8, 1040)
  - c. (16, 1024)
  - d. (8, 1024) ✓

Maximum occurs at  $t = -256/(2(-16)) = 8$  seconds.  $h(8) = -16(8)^2 + 256(8) = 1024$ . (8, 1024).

- 56 A rocket is launched with height modeled by  $h(t) = -16t^2 + 224t$ , where  $t$  is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).
- a. (8, 784)
  - b. (7, 784) ✓
  - c. (14, 784)
  - d. (7, 800)

Maximum occurs at  $t = -224/(2(-16)) = 7$  seconds.  $h(7) = -16(7)^2 + 224(7) = 784$ . (7, 784).

- 57 Solve  $x^2 - 14x + 42 = 0$  using the quadratic formula, giving exact answers.
- a.  $7 \pm \sqrt{7}$  ✓
  - b.  $7 \pm 2\sqrt{7}$
  - c.  $-7 \pm \sqrt{7}$
  - d.  $8 \pm \sqrt{7}$

$$x = (14 \pm \sqrt{(-14)^2 - 4(42)})/2 = (14 \pm \sqrt{28})/2 = 7 \pm \sqrt{7}.$$

- 58 Solve  $x^2 + 6x + 3 = 0$  using the quadratic formula, giving exact answers.
- a.  $-3 \pm 2\sqrt{6}$
  - b.  $-3 \pm \sqrt{6}$  ✓
  - c.  $3 \pm \sqrt{6}$
  - d.  $-2 \pm \sqrt{6}$

$$x = (-6 \pm \sqrt{(6)^2 - 4(3)})/2 = (-6 \pm \sqrt{24})/2 = -3 \pm \sqrt{6}.$$

59 Solve  $x^2 + 10x + 15 = 0$  using the quadratic formula, giving exact answers.

- a.  $-4 \pm \sqrt{10}$
- b.  $5 \pm \sqrt{10}$
- c.  $-5 \pm 2\sqrt{10}$
- d.  $-5 \pm \sqrt{10}$  ✓

$$x = (-10 \pm \sqrt{(10)^2 - 4(15)})/2 = (-10 \pm \sqrt{40})/2 = -5 \pm \sqrt{10}.$$

60 Solve  $x^2 - 8x + 3 = 0$  using the quadratic formula, giving exact answers.

- a.  $5 \pm \sqrt{13}$
- b.  $4 \pm \sqrt{13}$  ✓
- c.  $-4 \pm \sqrt{13}$
- d.  $4 \pm 2\sqrt{13}$

$$x = (8 \pm \sqrt{(-8)^2 - 4(3)})/2 = (8 \pm \sqrt{52})/2 = 4 \pm \sqrt{13}.$$

61 Solve  $x^2 - 2x - 17 = 0$  using the quadratic formula, giving exact answers.

- a.  $-1 \pm 3\sqrt{2}$
- b.  $1 \pm 4\sqrt{2}$
- c.  $1 \pm 3\sqrt{2}$  ✓
- d.  $2 \pm 3\sqrt{2}$

$$x = (2 \pm \sqrt{(-2)^2 - 4(-17)})/2 = (2 \pm \sqrt{72})/2 = 1 \pm 3\sqrt{2}.$$

- 62 A parabola has zeros at  $x = 2$  and  $x = -5$ , and passes through  $(-3, -30)$ . Find the leading coefficient  $a$  in its factored form  $y = a(x - (2))(x - (-5))$ .

a. 3 ✓

b. 4

c. 2

d. -3

Substitute  $(-3, -30)$ :  $-30 = a(-5)(2)$ , so  $a = 3$ .

- 63 A parabola has zeros at  $x = 3$  and  $x = -2$ , and passes through  $(-6, -36)$ . Find the leading coefficient  $a$  in its factored form  $y = a(x - (3))(x - (-2))$ .

a. -2

b. -1 ✓

c. 1

d. 0

Substitute  $(-6, -36)$ :  $-36 = a(-9)(-4)$ , so  $a = -1$ .

- 64 A parabola has zeros at  $x = -6$  and  $x = 3$ , and passes through  $(6, 108)$ . Find the leading coefficient  $a$  in its factored form  $y = a(x - (-6))(x - (3))$ .

a. 3 ✓

b. 2

c. -3

d. 4

Substitute  $(6, 108)$ :  $108 = a(12)(3)$ , so  $a = 3$ .

- 65 A parabola has zeros at  $x = -2$  and  $x = 6$ , and passes through  $(2, 32)$ . Find the leading coefficient  $a$  in its factored form  $y = a(x - (-2))(x - (6))$ .

a.  $-2$  ✓

b.  $2$

c.  $-1$

d.  $-3$

Substitute  $(2, 32)$ :  $32 = a(4)(-4)$ , so  $a = -2$ .

- 66 A parabola has zeros at  $x = -1$  and  $x = -2$ , and passes through  $(-8, -84)$ . Find the leading coefficient  $a$  in its factored form  $y = a(x - (-1))(x - (-2))$ .

a.  $2$

b.  $-3$

c.  $-2$  ✓

d.  $-1$

Substitute  $(-8, -84)$ :  $-84 = a(-7)(-6)$ , so  $a = -2$ .